PAVITHRA KUNE
COMET FWC 69

GATE EC 2009 Question 36

Question

If $X = 1$ in the logic equation

$$\left[X + Z\left\{\overline{Y} + \left(\overline{Z} + X\overline{Y}\right)\right\}\right]\left[\left\{\overline{X} + \overline{Z}(X+Y)\right\}\right] = 1$$

then

(A) $Y = Z$

(B) $Y = \overline{Z}$

(C) $Z = 1$

(D) $Z = 0$

Solution

Given:

$$X = 1$$

Substitute $X = 1$ into the equation:

$$\begin{gathered}\left[1 + Z\left\{\overline{Y} + \left(\overline{Z} + 1 \cdot \overline{Y}\right)\right\}\right] \\ \left[\left\{\overline{1} + \overline{Z}(1+Y)\right\}\right] = 1\end{gathered}$$

Since

$$1 + A = 1$$

the first bracket becomes

Now simplify the second bracket:

$$\overline{1} + \overline{Z}(1 + Y)$$

Since

$$\overline{1} = 0$$

and

$$1 + Y = 1$$

we get

$$0 + \overline{Z}(1)$$

$$\overline{Z}$$

Therefore the equation becomes

$$1 \cdot \overline{Z} = 1$$

Hence,

$$\overline{Z} = 1$$

So,

$$Z = 0$$

Final Answer

$$\boxed{Z = 0}$$

Therefore option **(D)** is correct.